

Day 3 Beginner-Friendly Full Slide Content

Course metadata

Field	Value
Course	Vinci AI Unitree Robotics Course
Day	Day 3
Deck	Beginner-Friendly Full Slide Content
Revised Topic Boundary	B2 platform hardware, industrial navigation/control subsystems, safe movement execution, manual override, gait adjustment, and Unitree B2 SDK point-to-point scripting only.
Excluded Content	Prior general communication-stack material, unrelated inspection reporting, SLAM, costmaps, and Day 4 field analytics.
Audience	Students new to Unitree B2 industrial quadruped operation and basic SDK-controlled movement.
Core Flow	Inspect → Configure → Command → Override → Observe → Validate
Branding	Vinci AI pale blue-gray background, dark navy text, teal-blue rules, top-right icon, diagonal watermark, and bottom-left copyright tag.

Teaching design notes

This revised Day 3 deck is intentionally scoped to **Unitree B2-exclusive beginner operation**. It removes broad middleware and general communication theory, then teaches the B2 as an industrial quadruped with real hardware limits, heavy-duty payload behavior, supervised movement, manual override discipline, gait awareness, and basic SDK-controlled point-to-point motion.

Visual and branding guidance: Use a pale blue-gray background, dark navy text, teal-blue rule systems, a small top-right Vinci AI icon/logo, a subtle diagonal watermark reading `Property of Vinci AI – Do Not Distribute`, and the bottom-left copyright tag `© 2026 Vinci AI. All rights reserved.`

The teaching pattern is **Inspect** → **Configure** → **Command** → **Override** → **Observe** → **Validate**. Each content slide uses one practical B2 idea, short slide text, instructor notes, and a diagram cue so the final presentation can be visual rather than text-heavy.

Section summary table

Section	Slides	Scope commitment
Scope and Safety Orientation	Slides 1–4	Defines the revised B2-only boundary and the safe motion loop.
B2 Platform Hardware	Slides 5–13	Covers hardware features, payloads, sensors, power, terrain limits, ports, and field safety.
Navigation & Control Subsystems	Slides 14–22	Explains B2-specific sensing, state, modes, gait, control layers, and readiness gates.
Basic Movements	Slides 23–31	Teaches control vectors, short motion tests, manual override, gait evidence, and movement validation.
Unitree B2 SDK Scripts	Slides 32–38	Introduces C++/Python SDK use, safe script structure, and segmented point-to-point logic.
Lab Validation	Slides 39–44	Runs one short B2 route, captures evidence, and validates B2-specific outcomes.

Cover

Day 3 Beginner-Friendly Full Slide Content

Unitree B2 Platform, Safety, Control, Movement, and SDK Point-to-Point Scripts

Vinci AI Unitree Robotics Course

Slide 1: Day 3 Is B2-First

Section: Scope

Purpose: Reset the deck around B2-exclusive learning outcomes and remove unrelated communication theory.

On-slide text:

- **Today:** B2 hardware, safety, movement, control subsystems, SDK scripts.
- **Not today:** broad middleware theory or unrelated robot software stacks.
- **Teaching flow:** Inspect → Configure → Command → Override → Observe → Validate.

Speaker notes: Begin by making the scope explicit. Students are not learning a general robotics software survey today. They are learning how the B2 behaves as an industrial quadruped, how safe commands are formed, how manual override stays ready, and how a basic script moves from one point to another.

Diagram cue: Two-column scope card with “B2 only today” on the left and “excluded topics” on the right; bottom arrow flow Inspect → Configure → Command → Override → Observe → Validate.

Slide 2: B2 Is Industrial Hardware

Section: Scope

Purpose: Frame the B2 as a heavy-duty industrial quadruped rather than a classroom toy.

On-slide text:

- **Approximate mass:** 60 kg including battery.[1](#)
- **Industrial emphasis:** payload, terrain ability, endurance, protection.
- **Student rule:** treat every command as real machine motion.

Speaker notes: Unitree lists the B2 at approximately 60 kg including battery, with industrial-grade specifications such as payload capability, terrain performance, and IP67 protection.[1](#) The teaching point is simple: beginners must respect the physical machine before they write code, because even a small command can move a large robot.

Diagram cue: Layered hardware silhouette with labels for mass, battery, legs, sensors, payload zone, and safety perimeter.

Slide 3: Control Starts With Respect

Section: Scope

Purpose: Connect safety mindset to every movement and script in the day.

On-slide text:

- **Before code:** clear area, operator ready, battery checked.
- **During motion:** small command, short duration, visible robot.

- **After motion:** stop, observe state, record evidence.

Speaker notes: A B2-specific lesson should start with operational respect. Students should practice low-risk behavior before discussing autonomy. The instructor should repeat that code is not separate from the robot; code becomes motion, motion changes balance, and balance can affect people, payloads, and the floor around the robot.

Diagram cue: Safety triangle: operator, robot, clear zone; arrows show before, during, and after movement checks.

Slide 4: One Safe Motion Loop

Section: Scope

Purpose: Introduce the repeated loop that replaces abstract communication theory.

On-slide text:

- **Loop:** Inspect → Configure → Command → Override → Observe → Validate.
- **Beginner goal:** prove one small motion before chaining commands.
- **Evidence:** command values, robot response, operator observation.

Speaker notes: The whole day can be taught through one repeatable B2 motion loop. Students inspect the robot, configure a safe mode, command a small movement, keep override ready, observe actual response, and validate whether the robot did what was intended. This makes the deck practical and non-redundant.

Diagram cue: Circular flow diagram with six stations labeled Inspect → Configure → Command → Override → Observe → Validate; include “small motion first” in the center.

Slide 5: Section 1 — B2 Platform Hardware

Section 1 — B2 Platform Hardware

Heavy-duty features, payload readiness, sensors, power, and field-safe handling.

Slide 6: Know the Body

Section: B2 Platform Hardware

Purpose: Identify B2 hardware zones students must recognize before operation.

On-slide text:

- **Trunk:** computing, battery, payload mounting, external ports.
- **Legs:** 12 degrees of freedom across four limbs.[1](#)
- **Head/sensing:** LiDAR and camera perception options.[1](#)

Speaker notes: Students should be able to point to the main B2 zones before touching software. The B2 development guide describes four legs with three joints each, giving 12 degrees of freedom, and lists perception components such as LiDAR, depth cameras, and optical cameras depending on configuration.[1](#)

Diagram cue: Exploded B2 body map with trunk, four legs, joint labels, sensor head, payload top, and battery area.

Slide 7: Payload Changes Behavior

Section: B2 Platform Hardware

Purpose: Explain why heavy-duty payload capability affects motion planning and safety.

On-slide text:

- **Standing load:** Unitree advertises at least 120 kg.[2](#)
- **Continuous walking load:** more than 40 kg.[2](#)
- **Rule:** heavier load means slower, smoother commands.

Speaker notes: The B2 is marketed for heavy-duty industrial use, including standing load and continuous walking load specifications.[2](#) Beginners should not interpret payload as permission to move aggressively. A payload changes inertia, balance, stopping distance, and how conservatively the operator should tune velocity and gait choices.

Diagram cue: Comparison table: no payload, light payload, heavy payload; rows for speed, turning, stopping, and caution level.

Slide 8: Sensors Support Awareness

Section: B2 Platform Hardware

Purpose: Introduce B2 sensing as practical awareness for navigation and control.

On-slide text:

- **3D LiDAR:** terrain feature awareness.[1](#)
- **Depth cameras:** near-field shape and obstacle cues.[1](#)
- **Optical cameras:** visual context and operator review.[1](#)

Speaker notes: The B2 guide lists an omnidirectional LiDAR, depth cameras, and optical cameras in typical perception configurations.¹ Students do not need advanced perception theory today. They need the practical meaning: sensors help the robot and operator understand the surrounding space before and during movement.

Diagram cue: Sensor fan diagram: LiDAR ring, front/rear depth cones, optical camera rectangles, and a simple obstacle outline.

Slide 9: Ports Require Discipline

Section: B2 Platform Hardware

Purpose: Teach safe handling of B2 external interfaces without drifting into general theory.

On-slide text:

- **External interfaces:** Ethernet, USB, 12 V, 24 V, and battery connections.¹
- **Critical rule:** no hot swapping aviation plugs.¹
- **Evidence:** inspect connection condition before powering payloads.

Speaker notes: Unitree explicitly warns that hot swapping aviation plug interfaces is strictly prohibited and may cause equipment failure not covered by warranty.¹ This is a practical B2 lesson: students should treat physical connectors as safety-critical items, not casual accessories.

Diagram cue: Checklist board with port types on the left and “power off, inspect, connect, strain-relieve, confirm” on the right.

Slide 10: Battery Is a Mission Limit

Section: B2 Platform Hardware

Purpose: Connect B2 battery capacity and runtime to safe lesson planning.

On-slide text:

- **Battery model:** BT2-10 with 45 Ah capacity.¹
- **Operating time:** approximately 4–6 hours depending on use.¹
- **Class rule:** plan commands inside the battery window.

Speaker notes: The B2 development guide lists the BT2-10 battery, 45 Ah capacity, and an operating time range of approximately four to six hours.¹ Students should connect battery status

to safe behavior. Low power can shorten experiments, limit retries, and require a controlled stop rather than rushed commands.

Diagram cue: Battery timeline with start check, mid-lab check, stop threshold, and recharge/replace decision.

Slide 11: Terrain Ability Is Not Permission

Section: B2 Platform Hardware

Purpose: Set expectations for terrain specifications and safe beginner limits.

On-slide text:

- **Slope ability:** greater than 45 degrees is listed under good conditions.[1](#)
- **Step height:** 20–25 cm is listed in specifications.[1](#)
- **Beginner rule:** test flat ground before challenging terrain.

Speaker notes: The B2 is specified for strong terrain performance, including slope and step abilities under stated conditions.[1](#) Beginners should not start at the edge of the specification. The safe teaching pattern is flat ground first, then small obstacles, then more complex terrain only with supervision.

Diagram cue: Terrain ladder: flat floor → shallow ramp → small step → instructor-approved challenge; safety gate before each level.

Slide 12: Weather And Site Matter

Section: B2 Platform Hardware

Purpose: Explain environmental operating boundaries as part of field protocol.

On-slide text:

- **Operating temperature:** –20 °C to 55 °C under good weather conditions.[1](#)
- **Protection level:** IP67 is listed for the B2.[1](#)
- **Protocol:** site condition decides whether motion begins.

Speaker notes: Students often assume a rugged robot can operate anywhere. The B2 specifications include environmental limits and protection level, but the phrase under good weather conditions still matters.[1](#) The operator must inspect ground, weather, visibility, bystanders, and payload stability before movement.

Diagram cue: Field readiness matrix with rows for ground, weather, people, payload, temperature, and visibility.

Slide 13: Safety Field Protocols

Section: B2 Platform Hardware

Purpose: Summarize beginner field rules before navigation and movement sections.

On-slide text:

- **Clear zone:** no feet, hands, cables, or loose objects near legs.
- **Operator role:** one person commands, one person watches when possible.
- **Stop path:** know the manual override before every run.

Speaker notes: This slide makes safety operational instead of abstract. Students should rehearse who gives commands, who watches the robot, where the robot can move, where people stand, and how motion stops. A B2 lesson should never allow code execution before the field protocol is visible.

Diagram cue: Top-down map with robot start box, no-go zone, operator zone, observer zone, and emergency stop route.

Slide 14: Section 2 — Navigation & Control Subsystems

Section 2 — Navigation & Control Subsystems

What the B2 must sense, decide, stabilize, and report during industrial movement.

Slide 15: Control Has Layers

Section: Navigation & Control

Purpose: Give students a beginner mental model of B2 industrial control subsystems.

On-slide text:

- **Perception:** sensors describe nearby terrain and obstacles.
- **Motion service:** converts intent into stable body movement.
- **State feedback:** tells the operator what actually happened.

Speaker notes: This is not a middleware lesson. It is a B2 control lesson. Students should picture layers inside the robot: sensing, body stabilization, gait generation, command

acceptance, and status reporting. The important beginner idea is that a high-level command must pass through safety and control layers before legs move.

Diagram cue: Layered stack: perception inputs → motion service → gait/body controller → legs → state feedback arrow.

Slide 16: Navigation Begins With State

Section: Navigation & Control

Purpose: Explain that movement decisions require current robot condition.

On-slide text:

- **State examples:** body posture, battery, mode, velocity, joint condition.
- **Question:** is the robot ready to accept movement?
- **Rule:** read state before sending a command.

Speaker notes: A beginner point-to-point script should not start by moving. It should first ask whether the robot appears ready. State feedback helps students confirm battery, posture, mode, and recent response. If state looks wrong, motion should pause until the issue is understood.

Diagram cue: Readiness gate diagram: state check inputs feed a large “move allowed?” decision box.

Slide 17: Modes Protect The Robot

Section: Navigation & Control

Purpose: Introduce control modes as safe contexts for different kinds of commands.

On-slide text:

- **Standing:** posture is stabilized before travel.
- **Velocity motion:** small directional command over time.
- **Special motions:** instructor-only unless approved.

Speaker notes: The B2 SDK examples expose high-level status and control patterns, including stand up/down, velocity movement, attitude balance, trajectory following, and special motions through sample programs.⁴ Beginners should learn that mode selection changes what a command means and how much risk it carries.

Diagram cue: Mode cards arranged from low risk to higher risk: stand, small velocity, trajectory, special motion.

Slide 18: Perception Helps Control

Section: Navigation & Control

Purpose: Connect B2 sensors to navigation and safe movement without teaching advanced mapping.

On-slide text:

- **Sensor role:** provide terrain and obstacle cues.
- **Control role:** maintain stable motion despite imperfect ground.
- **Operator role:** do not outrun what the robot can perceive.

Speaker notes: Students should connect sensors to practical control decisions. The B2 may include LiDAR, depth cameras, and optical cameras, but today the lesson is not advanced autonomy. The lesson is that perception supports safer movement choices, and the operator should command at a pace suitable for the site.

Diagram cue: Simple top-down path with sensor zones in front and rear, with speed arrows shrinking near obstacles.

Slide 19: Gait Is Movement Style

Section: Navigation & Control

Purpose: Define gait adjustment in beginner-friendly terms.

On-slide text:

- **Gait:** the pattern of leg timing and body support.
- **Why it matters:** stability, speed, turning, and terrain response.
- **Beginner habit:** change one gait-related setting at a time.

Speaker notes: A gait is not just animation. It is the style of leg coordination that changes how the B2 supports its body during motion. Beginners should treat gait adjustments as controlled experiments: one change, one short run, one observation, one note.

Diagram cue: Four-leg timing strip with alternating support phases and a “change one setting” badge.

Slide 20: Industrial Line Means Duty

Section: Navigation & Control

Purpose: Explain what is more B2-specific than consumer quadruped lessons.

On-slide text:

- **More payload:** commands must respect inertia.
- **More endurance:** runs require battery and thermal discipline.
- **More consequence:** site protocol matters as much as code.

Speaker notes: This slide answers the user's emphasis on what is more exclusive in B2. Compared with a lighter classroom robot, the B2's industrial positioning makes payload, site safety, power planning, and controlled gait behavior central topics. The deck therefore focuses on the B2 as working equipment.

Diagram cue: Comparison table: lightweight learning robot versus B2 industrial platform, with rows for payload, site protocol, inertia, and operator discipline.

Slide 21: Failures Show In Layers

Section: Navigation & Control

Purpose: Teach beginner debugging using B2 control layers.

On-slide text:

- **If no movement:** check mode, command size, and safety gate.
- **If unstable:** reduce speed, simplify gait, remove payload risk.
- **If uncertain:** stop and read state before retrying.

Speaker notes: B2 debugging should remain practical. Students should not jump directly to code edits when the robot does not behave as expected. They should ask whether the robot is in the right mode, whether the command is too small or too large, whether terrain is appropriate, and whether the state feedback supports another attempt.

Diagram cue: Troubleshooting funnel: symptom → likely layer → safe check → retry or stop.

Slide 22: Navigation Gate Before Motion

Section: Navigation & Control

Purpose: Create a validation gate before moving from theory to commands.

On-slide text:

- **Gate 1:** robot standing and stable.

- **Gate 2:** operator has manual override ready.
- **Gate 3:** path is clear and command is small.

Speaker notes: This validation gate keeps the deck concise by merging safety, evidence, and debugging into one motion-readiness check. Students should not run movement scripts until the B2 is stable, override is known, the path is clear, and the first command is deliberately small.

Diagram cue: Three-gate pipeline: Stable posture → Override ready → Clear path; final output “movement allowed.”

Slide 23: Section 3 — Basic Movements

Section 3 — Basic Movements

Control vectors, manual override, gait adjustment, and short safe motion tests.

Slide 24: A Vector Describes Intent

Section: Basic Movements

Purpose: Explain control vectors in simple movement language.

On-slide text:

- **Forward/back:** positive or negative travel intent.
- **Sideways:** lateral intent when supported by the control mode.
- **Yaw:** turn intent around the body center.

Speaker notes: A control vector is a compact way to describe how the robot should move. Beginners can think of it as a small set of numbers for forward motion, side motion, and turning. The B2 controller then tries to turn that intent into coordinated leg movement.

Diagram cue: Vector diagram over a top-down robot: forward arrow, side arrow, yaw curved arrow.

Slide 25: Small Commands Teach More

Section: Basic Movements

Purpose: Prevent beginners from using large movements before understanding response.

On-slide text:

- **Start tiny:** short duration and low speed.

- **Observe:** did direction, turn, and stop match expectation?
- **Increase only after proof:** do not guess upward.

Speaker notes: A heavy industrial quadruped teaches best through small commands. Students can learn direction, response delay, balance, and stop behavior without creating unnecessary risk. A clean small command is better evidence than a dramatic movement that nobody can explain.

Diagram cue: Scale graphic from tiny command to larger command; only the first step is green until validated.

Slide 26: Manual Override Stays Active

Section: Basic Movements

Purpose: Make manual override a required part of every movement test.

On-slide text:

- **Override is not optional:** know the controller/app stop path.
- **Do not bury control:** the operator must be ready before code runs.
- **Practice:** stop a harmless motion before testing longer motion.

Speaker notes: Manual override is part of the movement system, not a backup afterthought. Students should physically rehearse how to stop motion while the command is still simple. If they cannot stop a short slow movement confidently, they are not ready for autonomous point-to-point scripts.

Diagram cue: Operator hand/controller icon connected to a red stop path that interrupts the command arrow.

Slide 27: Stand Before Travel

Section: Basic Movements

Purpose: Teach posture readiness before sending travel commands.

On-slide text:

- **Posture first:** robot should stand cleanly and settle.
- **State check:** body attitude and mode should look expected.
- **Then move:** travel command follows stable stance.

Speaker notes: A point-to-point run begins with posture. The Unitree SDK examples include high-level stand up/down and motion tests, which reinforces the practical order: establish a stable posture, confirm the state, then send a small travel command.[4](#)

Diagram cue: Three-step panel: stand up → settle and read state → send small vector.

Slide 28: Forward Motion Test

Section: Basic Movements

Purpose: Define the first controlled movement test.

On-slide text:

- **Command:** slow forward vector for a short time.
- **Observe:** straightness, balance, stop behavior.
- **Record:** command value, duration, response notes.

Speaker notes: The first movement test should be boring on purpose. A slow forward command lets beginners confirm that the B2 accepts motion, moves in the expected direction, and stops cleanly. The evidence is simple: command, duration, actual response, and whether manual override remained available.

Diagram cue: Top-down lane with start box, short forward arrow, stop box, and observation checklist.

Slide 29: Turn Motion Test

Section: Basic Movements

Purpose: Show how yaw commands are tested safely.

On-slide text:

- **Command:** small yaw value in place or near-place.
- **Watch:** foot placement, body rotation, available space.
- **Stop:** end before drift becomes confusing.

Speaker notes: Turning is useful but can confuse beginners because the body rotates while feet step. Keep the turn small, use a clear floor marker, and stop early. The question is not whether the robot can spin dramatically; it is whether students understand command direction and observed yaw response.

Diagram cue: Top-down robot with curved yaw arrow, floor orientation marker, and stop boundary circle.

Slide 30: Gait Adjustments Need Evidence

Section: Basic Movements

Purpose: Teach gait changes as controlled beginner experiments.

On-slide text:

- **Change one thing:** gait style, speed, or terrain—not all three.
- **Compare:** before and after stability.
- **Document:** setting, surface, payload, and observed behavior.

Speaker notes: Gait adjustment becomes unsafe when students change multiple variables at once. The B2-focused habit is scientific but simple: keep the surface and payload fixed, change one gait-related setting, run a short test, and compare stability and operator confidence.

Diagram cue: Before/after comparison table with columns for setting, surface, payload, stability, and decision.

Slide 31: Movement Validation Gate

Section: Basic Movements

Purpose: Provide the required gate before progressing to scripted autonomy.

On-slide text:

- **Pass if:** stand, forward, turn, and stop are repeatable.
- **Hold if:** drift, delay, unstable gait, or uncertain stop occurs.
- **Evidence:** notes, values, state snapshot, operator sign-off.

Speaker notes: This gate merges debugging and evidence habits into the movement section. Students only progress after the B2 can stand, move forward, turn, and stop in a repeatable way. If the basic movement evidence is weak, the correct action is to simplify, not automate.

Diagram cue: Validation checklist with pass/hold branches; hold branch loops back to smaller commands.

Slide 32: Section 4 — Unitree B2 SDK Scripts

Section 4 — Unitree B2 SDK Scripts

Languages, state reading, basic command structure, and point-to-point autonomy.

Slide 33: SDK Is The Script Bridge

Section: Unitree B2 SDK

Purpose: Explain the Unitree SDK as the practical bridge between code and B2 behavior.

On-slide text:

- **C++ SDK:** `unitree_sdk2` supports building robot applications.³
- **Python SDK:** `unitree_sdk2_python` provides Python interfaces.⁴
- **Beginner focus:** read state, send small commands, stop safely.

Speaker notes: The official Unitree repositories describe SDK version 2 and a Python interface for SDK2.³ For this course, students only need a safe beginner subset: initialize the SDK, read status, command small movements, monitor response, and stop cleanly.

Diagram cue: Bridge diagram: student script → SDK call layer → B2 state/control interface → robot motion and feedback.

Slide 34: Choose The Language

Section: Unitree B2 SDK

Purpose: Clarify when students may use C++ or Python examples.

On-slide text:

- **Python:** faster for beginner experiments and readable scripts.
- **C++:** closer to compiled application workflows.
- **Course rule:** same safety logic regardless of language.

Speaker notes: The language choice should not change the operational discipline. Python may be easier for first scripts, while C++ may match production-style examples. In both cases, the safe script pattern stays the same: check state, command small motion, observe, stop, and log evidence.

Diagram cue: Comparison table: Python and C++ columns; rows for learning speed, application style, and shared safety rules.

Slide 35: Script Shape Matters

Section: Unitree B2 SDK

Purpose: Teach the skeleton of a safe point-to-point script.

On-slide text:

- **Initialize:** prepare SDK connection and parameters.
- **Read:** confirm robot state and mode.
- **Command:** send short vector steps toward the target.
- **Stop:** end with controlled zero-motion command.

Speaker notes: A beginner autonomous script should look predictable. It should not jump from launch to travel. The student should be able to point to the initialization block, state check, command loop, stop command, and evidence log. This makes debugging possible when the robot does not behave as expected.

Diagram cue: Code-structure pipeline: initialize → read state → command loop → stop → save log.

Slide 36: Point-To-Point Is Segmented

Section: Unitree B2 SDK

Purpose: Explain point-to-point movement as short supervised segments.

On-slide text:

- **Target:** a simple nearby point, not a long mission.
- **Segment:** move a little, observe, correct, continue.
- **Abort:** stop if state or site condition becomes unclear.

Speaker notes: Point-to-point does not mean the B2 blindly travels across a room. For beginners, it means a short route divided into safe motion segments. Each segment checks whether the robot is still stable, still inside the site boundary, and still moving toward the intended target.

Diagram cue: Top-down path map split into three short segments, each with observe and continue/abort decision nodes.

Slide 37: Use Pseudocode First

Section: Unitree B2 SDK

Purpose: Provide a beginner-readable script model without overloading syntax.

On-slide text:

- **Read state.** If not ready, stop.
- **For each segment:** send small vector, wait briefly, read state.
- **At target:** send stop command and save evidence.

Speaker notes: Students should understand the algorithm before copying syntax. Pseudocode makes the safety structure visible: readiness check, segmented command, repeated observation, stop command, and evidence saving. Once that logic is clear, language-specific examples become much easier to understand.

Diagram cue: Pseudocode flowchart with readiness diamond, segment loop, stop box, and evidence folder output.

Slide 38: Debug The Script Safely

Section: Unitree B2 SDK

Purpose: Create a focused debugging habit for SDK-controlled motion.

On-slide text:

- **No response:** check mode, state readiness, command size.
- **Wrong direction:** verify vector sign and frame assumption.
- **Unstable response:** stop, reduce speed, simplify gait.

Speaker notes: Debugging a B2 script should never begin by increasing command values. If the robot does not respond, first check readiness and mode. If direction is wrong, inspect the vector signs. If stability is poor, stop and simplify. Debugging is a safety process, not trial-and-error motion.

Diagram cue: Debug decision tree with three symptoms leading to safe checks and a final stop/retry decision.

Slide 39: Section 5 — Lab Validation

Section 5 — Lab Validation

Run the B2 safely, capture evidence, and prove the point-to-point workflow.

Slide 40: Pre-Run Checklist

Section: Lab Validation

Purpose: Give students a concise checklist before executing B2 movement scripts.

On-slide text:

- **Robot:** battery, stance, payload, ports, clear legs.
- **Site:** floor, boundary, observer, no loose cables.
- **Script:** small target, stop command, log folder ready.

Speaker notes: This checklist combines platform knowledge, safety protocol, and script readiness. It keeps Day 3 focused on B2 movement rather than broad theory. Students should not run point-to-point code until the robot, site, and script all pass the pre-run check.

Diagram cue: Three-column checklist labeled robot, site, and script; each column ends in a green “ready” box.

Slide 41: Run One Short Route

Section: Lab Validation

Purpose: Define the final beginner B2 lab run.

On-slide text:

- **Route:** start point to one nearby target.
- **Movement:** stand, move forward, adjust heading, stop.
- **Supervision:** manual override remains ready for the full run.

Speaker notes: The final lab should be short enough that students can explain every action. A good run includes a stable stand, a small forward segment, a heading adjustment if needed, and a clean stop. The point is not distance; it is controlled B2 behavior.

Diagram cue: Top-down route card with start, target, one heading correction, and manual override icon beside the path.

Slide 42: Evidence Proves Control

Section: Lab Validation

Purpose: Show what evidence students must keep after the B2 run.

On-slide text:

- **Keep:** command values, timestamps, state readings, operator notes.
- **Compare:** intended movement versus observed movement.
- **Claim:** only say success when evidence supports it.

Speaker notes: Evidence habits belong inside the B2 workflow. A student should be able to show what command was sent, when it was sent, what state was observed, and how the robot moved. This prevents vague claims such as “it worked” when the run was not actually verified.

Diagram cue: Evidence folder diagram containing command log, state snapshot, observation notes, and route sketch.

Slide 43: B2-Specific Outcome Map

Section: Lab Validation

Purpose: Tie every slide section back to the revised B2-exclusive outcomes.

On-slide text:

- **Platform:** hardware, payloads, safety protocols.
- **Subsystems:** sensing, state, modes, gait, control layers.
- **Movement:** vectors, override, gait adjustments.
- **SDK:** C++/Python basics and point-to-point scripts.

Speaker notes: This slide confirms that the revised deck is not a general communication deck. It is a B2-specific beginner deck. Each required outcome is covered directly: platform hardware, navigation and control subsystems, basic movements, and SDK-based point-to-point autonomy.

Diagram cue: Outcome coverage matrix with four rows and checkmarks against the five deck sections.

Slide 44: Exit Ticket: Prove B2 Readiness

Section: Lab Validation

Purpose: End with a concise student deliverable for the revised Day 3 scope.

On-slide text:

- **Explain:** how the B2 hardware changes safety decisions.

- **Demonstrate:** Inspect → Configure → Command → Override → Observe → Validate.
- **Submit:** script outline, run evidence, and safety reflection.

Speaker notes: The exit ticket asks for practical proof, not memorization. Students should explain one B2 hardware feature, describe a navigation/control subsystem, demonstrate a safe small movement or script outline, and submit evidence that the command, observation, and validation steps were completed.

Diagram cue: Exit ticket card with three boxes: explain, demonstrate, submit; bottom band states “B2 readiness is proven with evidence.”

Validation summary

Check	Result	Notes
Total slide items	45	One cover plus 44 numbered slides, within the requested 40–60 range.
Cover slide	Pass	## Cover contains only title, subtitle, and course information.
Section transition slides	Pass	Slides 5, 14, 23, 32, and 39 are minimal transition slides.
Required content-slide fields	Pass	Every non-transition content slide includes Section, Purpose, On-slide text, Speaker notes, and Diagram cue.
Diagram cues	Pass	Every content slide includes a visual cue using layered boxes, arrows, checklists, top-down maps, tables, or pipelines.
Removed excluded topics	Pass	The revised deck does not teach prior general communication-stack material or unrelated Day 4 topics.
B2 specificity	Pass	The deck focuses on B2 hardware, industrial control behavior, movement safety, gait adjustment, SDK languages, and point-to-point scripting.

Outcome coverage evaluation

Required revised Day 3 outcome	Coverage
Introduction to the B2 Platform: hardware features, heavy-duty payloads, and strict safety field protocols	100%
Understanding the Navigation & Control Subsystems of Unitree's industrial line	100%
Basic Movements: execution of control vectors, manual overriding, and gait adjustments	100%
Introduction to Unitree B2 SDK languages: writing basic autonomous point-to-point scripts	100%

References